

Challenge (Habanero)

Q1. Solve the inequality $2x - 5 > 11$.

- (A) $x > 8$
- (B) $x < 8$
- (C) $x > 4$
- (D) $x < 4$
- (E) $x = 8$

Q2. Graph the solution to the inequality $x + 2 \leq 7$ on a number line.

- (A) $x \leq 5$
- (B) $x \geq 5$
- (C) $x < 5$
- (D) $x > 5$
- (E) $x = 5$

Q3. A recipe for making cookies requires $\frac{3}{4}$ cup of sugar per batch. If you want to make at least 5 batches, how much sugar will you need in total?

- (A) > 3.5 cups
- (B) < 3.5 cups
- (C) $= 3.5$ cups
- (D) ≥ 3.5 cups
- (E) ≤ 3.5 cups

Q4. Write the solution to the inequality $x - 3 > 2$ in interval notation.

- (A) $(5, \infty)$
- (B) $(-\infty, 5)$
- (C) $[-5, 5]$
- (D) $(-5, 5)$
- (E) $[5, \infty)$

Q5. A construction project requires that the height of the build-

ing be at least 50 meters. If the current height is 40 meters, what is the minimum amount that needs to be added?

- (A) > 10 meters
- (B) < 10 meters
- (C) $= 10$ meters
- (D) ≥ 10 meters
- (E) ≤ 10 meters

Q6. The length of a room is at most 15 meters. If the width of the room is 8 meters, what is the maximum area of the room?

- (A) 120 square meters
- (B) 100 square meters
- (C) 80 square meters
- (D) 60 square meters
- (E) 40 square meters

Q7. Solve the inequality $x^2 - 4x - 3 > 0$.

- (A) $x < -1$ or $x > 5$
- (B) $x > -1$ and $x < 5$
- (C) $x < -1$ and $x > 5$
- (D) $x > -1$ or $x < 5$
- (E) $x = -1$ or $x = 5$

Q8. A water tank can hold at most 1000 liters of water. If 200 liters of water are already in the tank, what is the maximum amount of water that can still be added?

- (A) > 800 liters
- (B) < 800 liters
- (C) $= 800$ liters
- (D) ≥ 800 liters
- (E) ≤ 800 liters

Open Ended Questions (FRQ)

Q9. A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. If they sell at least 30 more whole wheat loaves than white bread loaves, and they sell at most 150 white bread loaves, find the range of the number of whole wheat loaves they can sell. Show your work and explain your reasoning.

Q10. A builder is constructing a building with a triangular roof. The length of the base of the roof is at least 10 meters, and the height of the roof is at most 15 meters. If the area of the roof is > 50 square meters, find the range of possible values for the base and height of the roof. Use inequalities to represent your solutions and graph them on a coordinate plane.

Chef's Tips

- Tip 1: Encourage students to read the questions carefully and identify the key elements of the inequality.
- Tip 2: Remind students to check their work and verify their solutions.
- Tip 3: Emphasize the importance of using proper notation and formatting when writing solutions in interval notation.